

Day 9, September 7 & 8, Unit 1

Agenda:

1. Spatial & Map Concepts Quiz
2. 1.2 & 1.3 Geo. Data Notes
3. Work on DBQ

HW: study for reading quiz on 9/13
(A) and 9/14 (B)

*Cell phones &
Smartwatches MUST go in
the holder!*

College Board Essential Knowledge

Identify different methods of geographic data collection

- ◆ Data may be gathered in the field by organizations or by individuals.
- ◆ Geospatial technologies include **geographic information systems (GIS), satellite navigation systems, remote sensing, and online mapping and visualization.**
- ◆ Spatial information can come from written accounts in the form of field observations, media reports, travel narratives, policy documents, personal interviews, landscape analysis, and photographic interpretation.

Explain the geographical effects of decisions made using geographical information

- ◆ Geospatial and geographical data, including census data and satellite imagery, are **used at all scales for personal, business and organizational, and governmental decision making purposes.**



1.2 Geographic Data

Types of Data

Qualitative:

descriptive and based on field observations, open-ended interviews (**words**): subjective



Quantitative:

measurable/expressed in **numbers**; measurable, more objective



Geographers collect data in various ways/methods

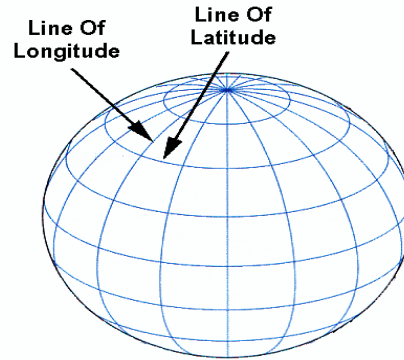
- ◆ Remote Sensing
- ◆ Global positioning system (GPS)
- ◆ Geographic information system (GIS)
- ◆ Census

Remote Sensing

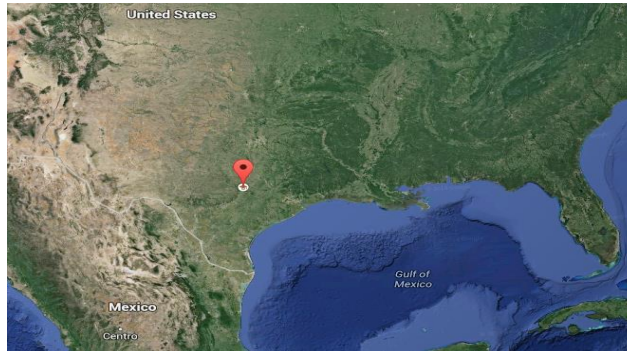
The acquisition of data about the Earth's surface from a satellite orbiting the planet or from other long distance methods (like planes or drones)



GPS



A system of satellites that use longitude and latitude to pinpoint absolute location



GIS

A computer software system that stores, organizes, analyzes, and displays geospatial data to observe geo. patterns, processes (creates multi-layered maps & other forms of geovisualization)

- ◆ analyze patterns & processes and make predictions



GPS answers the
question...
Where is the nearest
Target?

GIS answers the question...
Where should we build the
next Target?

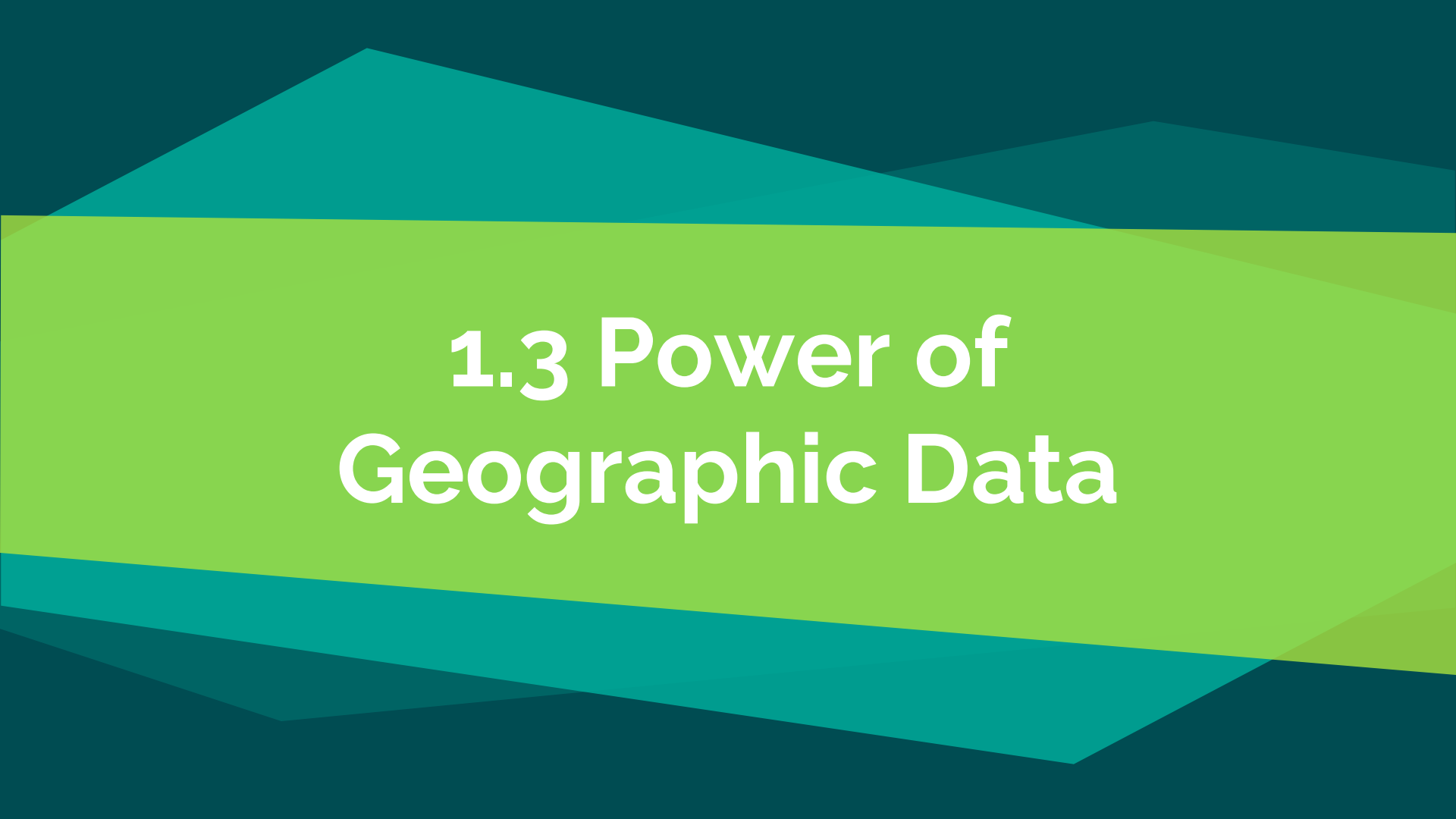
How we are using GPS &
GIS today!



Census Data



- ◆ Census data is collected every 10 years in the US
- ◆ Census data is used to determine where to build new schools, which services are needed, how much of those services are needed, the amount of funds allocated to an area, etc.
- ◆ Also used to redraw voting districts (important!)

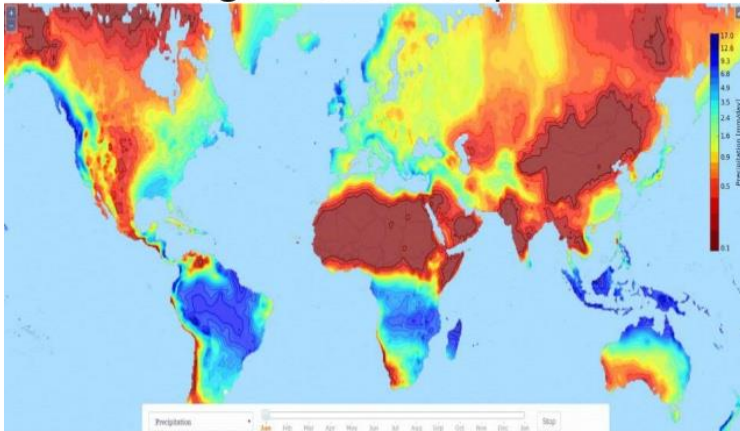


1.3 Power of Geographic Data

Geographic data helps us see SPATIAL PATTERNS

What are some conclusions that you can draw from the patterns on these maps?

Average Global Precipitation



Retrieved from: <http://www.ecoclimax.com/2016/08/climate-maps.html>

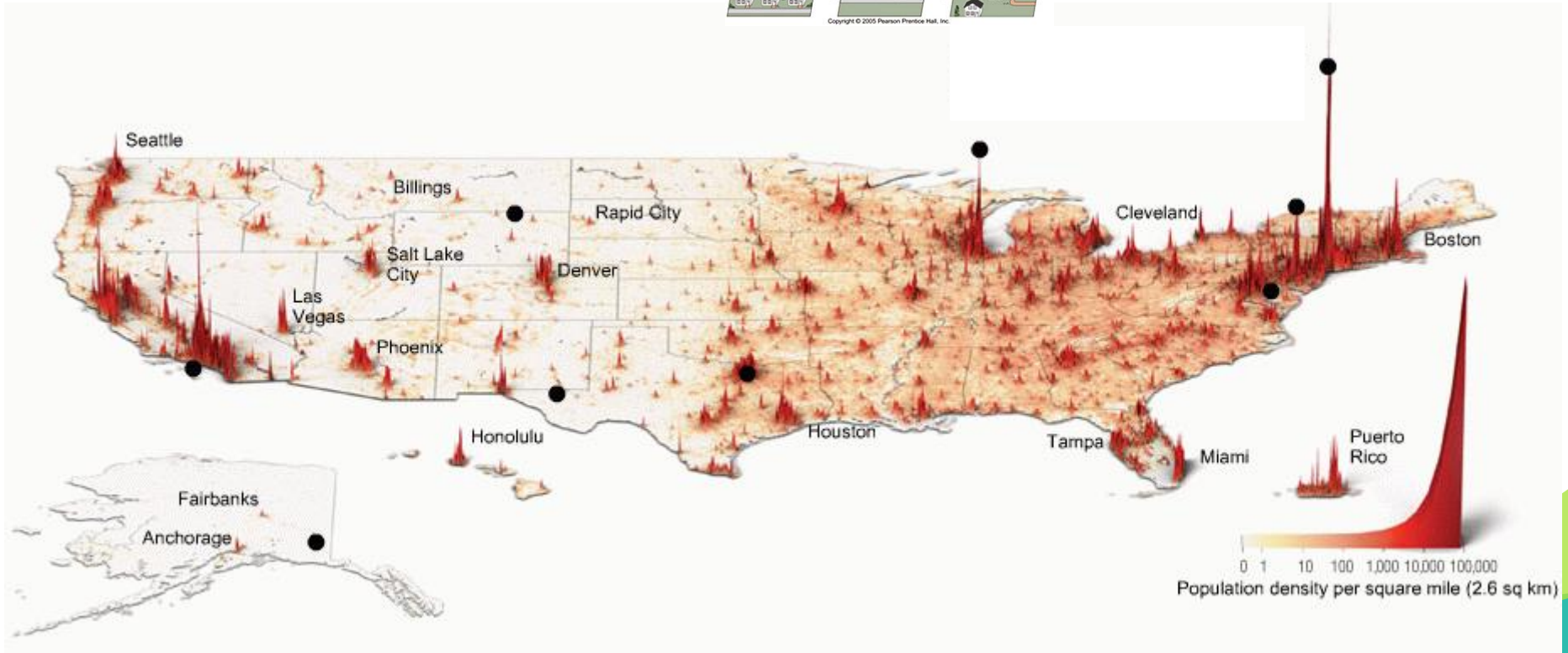
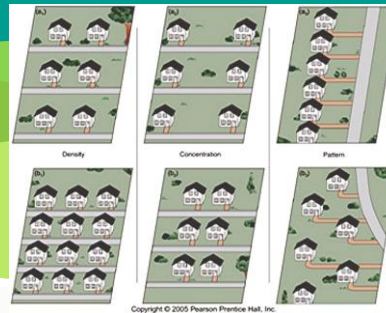


Spatial Patterns

What do you notice?



Spatial Distribution



Analyzing geographic data allows us to

Identify, describe, explain, compare, analyze:

- spatial patterns

- spatial distribution

- spatial interactions & relationships

How Data is Used?

- ◆ Helps us understand problems → make decisions → measure effectiveness
 - ◆ Examples
 - ◇ Public health officials planning for flu season
 - ◇ People looking for a home
 - ◇ Businesses looking to open up a new location
 - ◇ Local governments addressing homelessness/lack of affordable housing



USA demographics

from the Esri GeoInquiries™ collection for Human Geography

<https://education.maps.arcgis.com/home/webmap/viewer.html?webmap=570fdb49796243bd8a05fc6e1df4c417>

In summary...

